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RESEARCH ARTICLE

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The usability of a digital diary from the perspectives of intensive care patients' relatives: A pilot study

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Abstract

Background: Diaries have been used regularly in international settings as an evidence-based and easily applicable intervention following a person-centred approach in the intensive care unit (ICU). In addition, a diary web application known as 'Post-ICU' has been implemented.

Aim: To explore the usability of an innovative digital diary from the perspectives of intensive care patients' relatives.

Study Design: A cross-sectional online survey study was applied among a convenience sample of relatives in the ICUs of two hospitals in the Netherlands. The investigator-developed questionnaire included, among other things, items with the appreciation of the layout, user friendliness and functionality of the diary. Relevance and applicability were rated between 1 and 10. Data were analysed with SPSS© software, version 27, and reported as the means (±standard deviation [SD]) and percentages.

Results: Sixty-three relatives with an average age of 51 years (SD ± 14.3) participated in the study; there was a slight predominance of women (57%). All but one participant found using the digital diary easy and were able to upload photos to the diary. The participants had invited other relatives (75%) and nurses (61%) to write in the diary, which they viewed as easily feasible (89%). The relevance and applicability of the diary were rated with mean scores of 8.1 (SD ± 1.9) and 8.3 (SD ± 1.6), respectively.

Conclusions: The participants found the Post-ICU diary web application highly valuable and easily feasible. Perceived ease of use, perceived usefulness and technophobia were not found to influence the usability of the digital diary.

Relevance to Clinical Practice: The implementation of this new digital tool supports a person-centred ICU policy because of its focus on the personal diary entries of the patient and the collaborative writing process featuring relatives and professionals.

KEYWORDS
digital innovation, E-health, family, intensive care unit, PICS

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1 | INTRODUCTION

The topic of surviving a critical illness has attracted attention from the public and many health care professionals outside the intensive care unit (ICU). Admission to the ICU with a life-threatening illness and subsequent advanced medical treatment impacts both the patient and his or her entire family and support system.[1] Critical illness might lead to the onset of several health-related impairments among 50%–70% of ICU survivors in the physical, cognitive and psychological domains; these impairments are collectively known as post-intensive care syndrome (PICS)[2–4] (Figure 1). ICU patients' relatives (partners, children, parents and friends) are at risk of developing psychological distress, such as symptoms of post-traumatic stress disorder (PTSD) (35%–57%), anxiety (15%–25%) and depression (5%–36%).[5] Relatives of deceased ICU patients may experience long-lasting grief as a consequence of the unpredictable and burdensome situation they face.[6] These impairments are collectively referred to as PICS-Family (PICS-F)[7,8] (Figure 1).

The health model used in this study is based on a medical description of PICS and PICS-F, which was first created at an international stakeholder conference,[3] has been studied in further detail in ongoing scientific work[2,4,9–11] and has more recently been extended to the paediatric setting.[12] This model includes the importance of impairments in social health as a result of critical illness alongside the physical, cognitive and emotional domains that had previously been emphasized.

Undoubtedly, emotional distress because of critical illness and social outcomes are intricately intertwined.[4,10,12,13] This model recognizes the fact that the interrelationships between ICU survivors and their relatives are central to the trajectory of recovery. Additionally, relatives may reduce their social activities,[5] thus further increasing their risk of PICS-F. These health-related impairments, in turn, influence the outcomes of ICU survivors. Adequate communication and the continuous support of relatives can contribute both to the recovery of the patient and to their own mental well-being.

What is known about the topic

- Intensive care unit (ICU) diaries in hard copy (paper) version have been used regularly in various hospitals in international settings with positive effects in reducing health-related impairments in ICU survivors and their relatives.
- Innovative strategies that emerged during the COVID-19 pandemic illustrated new digital possibilities for providing information and supportive care.

What this paper adds

- ICU patients' relatives experienced a diary web application as highly valuable and easy to use.
- ICU professionals might suggest the use of a digital diary to relatives and encourage use by focusing on its feasibility.

2 | BACKGROUND

During the COVID-19 pandemic, the symptoms of PICS-F increased for several reasons, such as (1) the severity of the disease; (2) its high rates of morbidity and mortality; (3) the needed physical distancing and restrictions on visiting because of the need for infection control; and (4) the social impact of and media attention to ICU admission.[14–18] These unique challenges resulting from the pandemic affected the relatives of patients both with and without

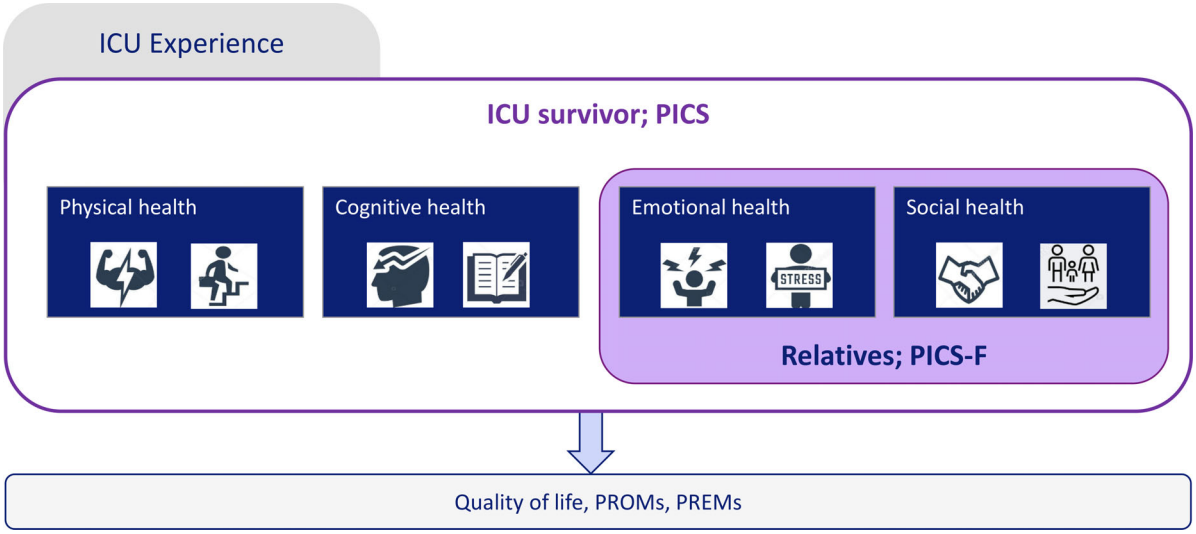

FIGURE 1 Model of post-intensive care syndrome (PICS) and PICS-family (PICS-F). ICU, intensive care unit; PREM, patient reported experience measure; PROM, patient reported outcome measure.

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COVID-19. During the years prior to the pandemic, several interdisciplinary rehabilitation initiatives were developed and implemented both globally[19,20] and in the Netherlands[21] to reduce and ideally prevent the symptoms of PICS/PICS-F. These initiatives, which have been suggested in an international guideline for application to family-centred care in the neonatal, paediatric and adult ICUs, ranged from the use of ICU diaries to the integration of multidisciplinary follow-up services.[22]

2.1 | ICU diaries

Diaries have been used regularly in international settings as an evidence-based and easily applicable intervention following a person-centred approach in the ICU.[23] For ICU survivors, reading a diary is an effective way of coping with the traumatic aftermath of critical illness. Systematic review studies including meta-analyses have shown that the use of a diary leads to a decrease in depression and anxiety in ICU survivors.[24–26] The opportunity to talk about what happened and reviewing pictures has been described as valuable and helpful for patients with respect to understanding their memories and delirious illusions (e.g., dreams, nightmares and hallucinations) during their stay in the ICU.[27–29] For relatives, access to tools that allow them to do something for their loved one represents a form of support in the coping process. Writing in a diary allows the individual to establish a timeline, express feelings and structure thoughts. To date, keeping a diary during an ICU stay and thereafter is the only evidence-based intervention that has exhibited a positive effect on psychological distress in relatives.[30,31] However, during the COVID-19 pandemic, hard copy diaries became less practical because of the absence or limited presence of relatives in the ICU.

2.2 | Digital innovation in ICU support

Innovative strategies that emerged during the pandemic illustrated the digital possibilities for providing information and supportive care. For example, web-based interventions, such as video conferencing and text messages, have been effective alternatives to facilitate interhuman contact, information and support.[32] A web application, known as the 'Post-ICU' diary, was introduced at the start of the first wave of COVID-19 in the Netherlands (March 2020). It was expected that this digital diary would have the potential to ameliorate the increasing frequency and intensity of PICS and PICS-F.[33] The utilization of the Post-ICU diary is described in Box 1.

A theoretical framework for the applicability and adoption of technology, such as the digital diary, is the Technology Acceptance Model (TAM).[34] Both perceived ease of use (PEOU) and perceived use (PU) influence users' behavioural intention.[35] Other aspects, such as users' personal characteristics and potential fear of using computers (e.g., low digital competences or reading ability) might have a direct influence on behavioural intentions.[36,37]

BOX 1 Post-ICU utilization

The Post-ICU diary is available from https://post-icu.com, and the screen layout is shown in Figure 2. Post-ICU can be used on every digital device with a screen (e.g., laptop, tablet smartphone), and special training to use the application is not needed. The patient or a legal representative (the first contact person in the electronic patient record) remains the owner of the digital diary and the included data (text entries, photo's, emoticons) at all times; the Post-ICU diary is compliant with all privacy and security requirements with certification and safe data storage. The keeper of the diary, which usually is managed by the first contact person, decides who may access to Post-ICU, and invites these individuals to write or share data entries. Several individuals related to one patient can read and collaboratively write in the digital diary. They can write entries in Post-ICU as many times as they wish, partners of the patient are most willing to be actively involved. They can also upload photos and continue writing after ICU discharge. ICU professionals can write short messages if they wish, ideally on a daily base, and upload photos to Post-ICU. Although nurses will use Post-ICU most frequently compared with other health care professionals, social workers and medical doctors are involved in its utilization as well. Successful implementation of Post-ICU is of influence on its utilization by all stakeholders. Possible facilitating and hindering factors, as well as tailed strategies for its implementation, have been described previously.[33] The diary entries are a valuable resource for ICU survivors and relatives, allowing them to describe the situation and circumstances they face, which can vary greatly from day to day.

3 | AIMS AND RESEARCH QUESTIONS

Hitherto, it has remained unknown how the various stakeholders appreciate the Post-ICU diary. The research question pursued in this study was: 'How do ICU patients' relatives evaluate the usability of the Post-ICU diary?'

Therefore, the aim of this study was to explore the usability of the digital diary from the perspectives of the relatives of ICU patients. In addition, PEOU, PU and technophobia related to the digital diary were explored as influencing factors in the uptake of an innovative digital communication tool.

4 | DESIGN AND METHODS

The overall study was conducted as a larger explorative multi-method pilot study prior to the national scale-up of the Post-ICU diary.[33]

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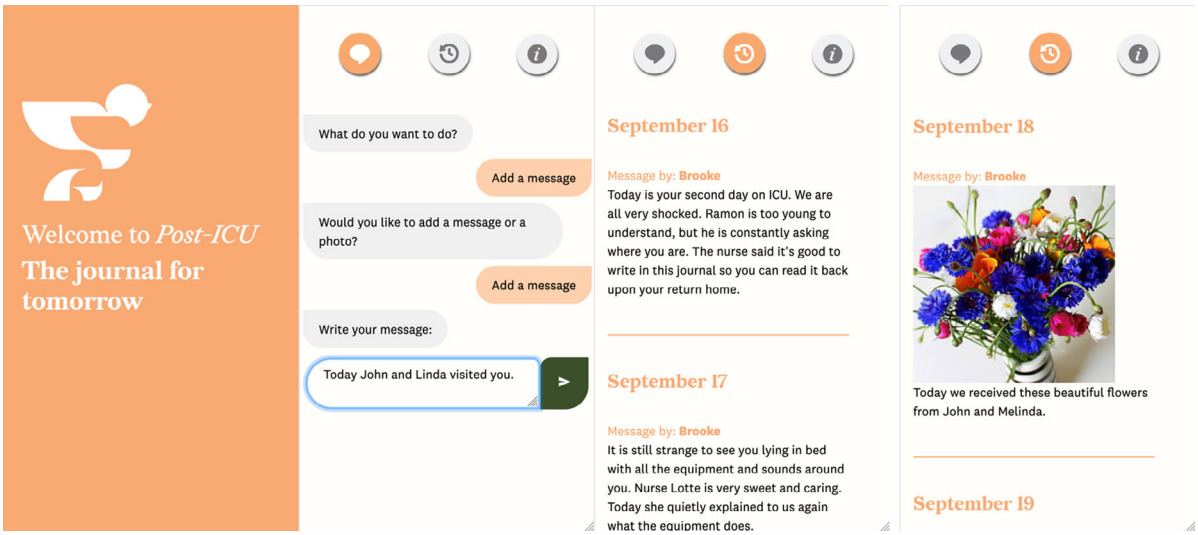

FIGURE 2 Overview of screen layout Post-ICU diary. ICU, intensive care unit. *Source*: Games for Health©.

This pilot was intended to pretest the study materials and procedures—by consulting with relevant stakeholders—to facilitate full evaluation in the future. The qualitative part of the study, which explored the perspectives of ICU nurses on the applicability of the digital diary, has been published previously.[33] The present study has a cross-sectional quantitative design exploring the perspectives of ICU survivors' relatives. The STROBE checklist served as a reporting guideline.[38]

4.1 | Setting

Contact persons in five hospitals in the Netherlands were invited to participate in the pilot study based on their geographical locations and personal relationships with the research group. All five agreed to participate, although because of a delay resulting from internal procedures and high workload during the COVID-19 pandemic, only two hospitals fit within the time frame of the present study. These ICUs, including one ICU at an academic hospital and one ICU at a tertiary teaching hospital, participated actively in this quantitative study. Experiences with the digital diary ranged from three to 8 months at study onset per ICU.

4.2 | Study sample and recruitment

The population consisted of a convenience sample of ICU patients' relatives recruited between September 2020 and May 2021. To be eligible to participate in this study, the relatives needed to meet all of the following criteria: relative of patient with an ICU stay longer than 48 h featuring mechanical ventilation; above 18 years of age; sufficient knowledge of the Dutch language (including the ability to discuss, read and understand relevant information regarding various aspects of the study); presence during the ICU stay; and legal responsibility at the time of admission, because of being the first contact person listed in the electronic patient record. Relatives of patients in unstable medical situations (such as a resuscitation setting or active bleeding problems) and end-of-life care were not eligible.

One researcher informed the relatives on the background of the study, the objectives and the relatives' expected contribution orally. If a relative was interested in participating in the study, contact details, including an e-mail address, were collected and additional written information material regarding the study was provided. Subsequently, a link to the digital questionnaire was sent to the participants. In addition, a QR code for the questionnaire was provided. At the start of the questionnaire, the participants signed informed consent.

4.3 | Ethics statement

The study was approved by the Medical Ethics Review Board of the Erasmus University Medical Center, Rotterdam, the Netherlands, which served as a coordinating centre (MEC-2020-0640). The Board decided that the rules laid down in the Dutch Medical Research Involving Human Subjects Act did not apply to this research proposal. The study was conducted according to the principles of the Declaration of Helsinki (64th WMA General Assembly, Fortaleza, Brazil, October 2013). Participation in this study was voluntary and participants could leave the study at any time they wished for any reason without consequences.

4.4 | Data collection tools

The tool used for this study was a questionnaire featuring 65 items in total. Demographic variables, such as relationship to the patient, sex, age and level of education, were plotted at the beginning. Next, an item followed to ask whether the Post-ICU diary had been used

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TABLE 1 Overview of the item scores on usability of the Post-ICU diary from the perspectives of study participants ($N = 44$[a]).

Survey item	Proportion (%)	Mean (±SD)
How did you manage to start the Post-ICU diary, (e.g., downloading and starting the application)?		4.16 (±0.65)
Neutral	13.6	
Easily feasible	56.8	
Very easily feasible	29.5	
How do you rate the ease of use of the Post-ICU diary in general?		3.98 (±0.66)
Difficult	2.3	
Neutral	15.9	
Easily feasible	63.6	
Very easy feasible	18.2	
How did you experience writing a text message in the Post-ICU diary?		4.14 (±0.74)
Difficult	2.3	
Neutral	13.6	
Easily feasible	52.3	
Very easy feasible	31.8	
How did you experience making and adding a photo in the Post-ICU diary?		3.93 (±0.82)
Very difficult	2.3	
Neutral	22.7	
Easily feasible	52.3	
Very easy feasible	22.7	
Have you invited other relatives to write in the Post-ICU diary?		
Yes	75.0	
No	25.0	
How did you experience inviting other relatives to the Post-ICU diary?		4.06 (±0.75)
Somewhat dissatisfying	6.1	
Neutral	6.1	
Satisfying	63.6	
Very satisfying	24.2	
Have you invited nurses to write in the Post-ICU diary?		
Yes	61.4	
No	38.6	
How did you experience inviting nurses to the Post-ICU diary?		4.07 (±0.99)
Not satisfying at all	3.7	
Somewhat dissatisfying	3.7	
Neutral	11.1	
Satisfying	44.4	
Very satisfying	37.0	
What do you think of the layout of the Post-ICU diary?		
Attractive	20.5	
Appropriate	77.3	
Disruptive	2.3	

(Continues)

TABLE 1 (Continued)

On which device did you use the Post-ICU diary the most?	
PC or laptop	40.9
Mobile phone	50.0
Tablet	9.1
Have you used the Post-ICU diary to talk to your loved one (here referred to as the patient) about the ICU admission?	
Yes	61.4
No	38.6
Have you used the Post-ICU diary to talk to health care providers (e.g., general practitioner) about the IC admission?	
Yes	9.1
No	90.9

Note: Answer categories in the Likert-type scales with no data are not shown in the table.
Abbreviation: SD, standard deviation.
[a]Participants who used and assessed the Post-ICU diary, which was 44/63.

(yes/no) and one item regarding the behavioural intention to continue using the diary (yes/no). The questionnaire furthermore contained an investigator-developed sub-scale to examine (1) usability and validated sub-scales to gain insight in (2) PEOU, (3) PU and (4) the impact of technophobia. Different answer categories (e.g., yes/no, five-point Likert scale, rating 1–10) were applied where most appropriate to the item.

4.4.1 | Usability

For this study, usability was defined in terms of the accessibility, relevance and applicability of the Post-ICU diary from the perspectives of the study participants. Based on the researchers' expertise, a questionnaire was developed on these aspects. Therefore, the 21-item questionnaire included, among others, questions regarding participants' experiences with the layout, user friendliness and functionality of the diary; these items were scored on a five-point Likert scale (1 = not important/not satisfying/very difficult; 5 = extremely important/satisfying/easily feasible). For relevance and applicability, a rating between 1 and 10 was requested (1 = very bad; 10 = very good). The measurement tool used to investigate the practicability of the Post-ICU diary had not been used previously, but the research team tested the items for comprehensibility via consultation with three volunteer family members.

4.4.2 | Perceived ease of use

Perceived ease of use (PEOU), which was defined as the degree to which a person believes that using technology requires little effort, was measured using six items[34] of a validated instrument.

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The researchers adapted a more general term, such as 'technologies', to 'digital diary' for ease of the respondents to suit the context of their experiences with the digital diary using the forward–backward translation procedure. The items, such as 'It is easy for me to learn to use the Post-ICU diary', were scored on a five-point Likert scale (1 = totally disagree; 5 = totally agree). Scale reliability (Cronbach's alpha = .94) demonstrated strong internal consistency.

4.4.3 | Perceived usefulness

PU, which was defined as the extent to which someone experiences the added value or usefulness of the technology, was measured using three items of a validated instrument.[39] The researchers adapted a more general term, such as 'technologies', to 'digital diary' for ease of the respondents to suit the context of their experiences with the digital diary using the forward–backward translation procedure. The items, such as 'It would be nice for me to use the Post-ICU diary', were scored on a five-point Likert scale (1 = totally disagree; 5 = totally agree). Scale reliability (Cronbach's alpha = .89) indicated strong internal consistency.

4.4.4 | Technophobia

Technophobia, which was defined as an irrational fear caused by the side effects of advanced technologies, was measured using 16 items of a validated instrument[40]; the researchers adapted a more general term, such as 'the system', into 'digital diary' for ease of the respondents to suit the context of their experiences with the digital diary using the forward–backward translation procedure. The items, such as 'I am afraid to use some functions of my mobile phone', were scored on a five-point Likert scale (1 = totally disagree; 5 = totally agree). Scale reliability (Cronbach's alpha = .94) indicated strong internal consistency.

At the end, two open questions invited participants to elaborate on their experiences with the Post-ICU diary: 'What other features would you like to add to the diary?' 'If you could change one thing about the diary, what would it be?' This final open question provided participants with the opportunity to add any desired comments or experiences with the Post-ICU diary.

4.5 | Data analysis

Data were analysed using SPSS© version 27 software. Categorical variables were computed as proportions; continuous variables were computed as medians and interquartile ranges (IQRs) or means and standard deviations (SDs), as appropriate. Normal distribution was checked using kurtosis and skewness. Relationships between variables were analysed using the Pearson's product–moment correlation for continuous variables or the Spearman rank-order correlation coefficient for ordinal variables. To test the mediating effect of PEOU and PU on the relationship between technophobia and intention to use the Post-ICU diary, the PROCESS macro was used. The PROCESS macro is essentially a modification to statistical programs such as SPSS and computes regression analyses containing various combinations of mediators, moderators and covariates.[41] A two-tailed *p*-value <.05 was considered to indicate a significant difference.[42]

5 | RESULTS

Sixty-three relatives with an average age of 51 years (SD ± 14.3) participated in the study. Most were a partner (52%) or son/daughter (25%) of the ICU patient. There was a slight predominance of women (57%) and highly educated participants (46% held a bachelor's or academic degree). All but one of the participants found using the digital diary to be easy and were able to upload photos. The participants invited other relatives (75%) and nurses (61%) to write in the diary, which was viewed as easily feasible (89%). The majority of participants reported that the diary had an attractive or appropriate layout (98%) (Table 1). The relevance and applicability of the diary were rated with mean scores of 8.0 (SD ± 1.9) and 8.4 (SD ± 1.6), respectively.

TABLE 2 Mean (*M*), standard deviation (SD), percentage (%) and correlations of technophobia, perceived ease of use, perceived use, intent of use and demographic variables (*N* = 63).

Variable	*M*	SD	%	1	2	3	4	5	6	7
1. Technophobia	1.89	0.63		1						
2. PEOU	3.94	0.80		−.12	1					
3. PU	3.99	0.85		.12	.73*	1				
4. Intent of use	4.11	1.00		−.01	.60*	.78*	1			
5. Age	51.29	14.25		.22	.02	.06	.02	1		
6. Sex[a]			57.1	.16	.08	.23	−.03	−.04	1	
7. Education[b]			46.0	−.17	−.14	−.13	.04	−.45*	−.23	1

Abbreviations: PEOU, perceived ease of use; PU, perceived use.
[a]Female.
[b]High education (bachelor or academic).
*$p < .01$.

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TABLE 3 Summary of participants' notes $N = 24$ (36.9%) who answered on the open questions.

Quotation	Respondent
• A very good initiative. Fortunately, my wife was only in the ICU for a few days. I can imagine that the diary is very valuable for patients and family.	Respondent 12
• I liked that there was the option to use the digital diary.	Respondent 16
• My sister can read it back if she needs to and it was nice for me to write in the diary. You are actually writing off your sorrows and worries at that moment. I also often kept it a bit light-hearted so that if my sister read it would not become too heavy for her.	Respondent 21
• It has been of great support to us to read the entries of my family, I felt connected in worrisome time. In addition, it was also a pass-time. You had something to do before going back to the hospital. It was a nice distraction as well.	Respondent 27
• Personally, I would have preferred a written diary.	Respondent 61

Technophobia was low ($M = 1.89$, SD ± 0.63) among both users ($n = 44$) and non-users ($n = 19$) of the Post-ICU diary. No significant relationship was found between technophobia and PEOU; ($t(60) = -0.78$, $p = .44$). PEOU and PU were measured as high; 3.94 (SD ± 0.80) and 3.99 (SD ± 0.85), respectively (Table 2). A difference in PEOU was found between users and non-users of the diary ($t(60) = 3.49$, $p < .001$), such that users experienced higher PEOU. Accordingly, people who used the diary indicated greater ease of use than people who did not use the diary. PEOU and PU exhibited a mutual positive correlation; however, no mediating effect was found with regard to the relationship between technophobia and intention to use the Post-ICU diary. The regression analysis found no relationship between PEOU and behavioural intention after controlling for PU and technophobia ($t(58) = -0.29$, $p = .770$). A positive relationship was found between using the diary and behavioural intention ($t(58) = 3.84$, $p < .001$), indicating that people who used the diary had greater intention to (continue to) use it than non-users of the diary.

Some participants answered the open questions. They would like to have the opportunity to upload video fragments, print the diary entries as a booklet and add emoticons. In addition, entries from other relatives and friends were appreciated as tokens of connectedness and social support. Overall, the participants expressed their positive experiences with the new tool (Table 3).

6 | DISCUSSION

The participants perceived the new digital diary to be feasible, as indicated by the fact that the majority of participants could easily begin using the diary, made use of its full functionality and identified it as a user-friendly and easily accessible new tool. Relevance and applicability were rated high (both above 8) and several notes were found underpinning the positive evaluations and recommendations to continue using the diary. Three out of every four participants had invited other relatives to exchange short messages and/or share personal stories, thus suggesting the need for social support and connectedness. Moreover, ICU professionals were invited to engage in 'small talk' regarding daily care and the circumstances of the patient. This was found to be satisfactory by 81.4% of the respondents. Therefore, the Post-ICU diary facilitated social interaction between the relative, his or her own network and the ICU team, which may be considered as social support. Social support entails that one has relatives to whom one can turn in times of need or crisis.[43] Social support enhances quality of life and serves as a buffer against adverse life events.[44] Indeed, it is a complex phenomenon that, however, cannot be fully explored among the users of the digital diary solely on the variables included in this study. However, the Post-ICU diary in itself may evoke social interaction among the relatives. This diary was intended for use by several persons, facilitating interaction and mutual support in situations of turmoil. Based on valuable interactions, it can be concluded that although the digital diary does not constitute a communication tool in its own right, it appears to play a role in facilitating connectedness and human interaction. The Post-ICU diary induces a sense of belonging to social groups, including a sense that one is part of a family. This sense was extremely important during the COVID-19 pandemic, when connection and social support were made difficult through quarantine measures. The fact that good communication with relatives, family participation and open visiting hours contributed to patient and family satisfaction became clearer.[45] The value of supportive communication and shared decision-making is inextricably related to the basic needs of ICU patients' relatives and may influence the prevalence of PICS-F.[46] As mentioned previously, the circumstances of the COVID-19 pandemic conflict with these needs.[47]

Participants did not resist using the new digital tool; technophobia was found low and did not limit the usability of Post-ICU diary. Furthermore, the participants believed that using the new digital technology needed little effort and added value, as was shown with high levels of PEOU and PU. The findings in this study were not supportive of TAM. A reason for this discrepancy might be that the response rates were too small and therefore did not provide sufficient power to identify significant differences in the behaviour associated with using the digital diary. Overall, the preliminary results of this study should be interpreted with caution, as they were obtained from a small convenience sample of ICU patients' relatives. Therefore, the gained insights require confirmation in a larger feasibility study.

6.1 | Lessons learned

The COVID-19 pandemic represented one of the greatest challenges for ICU teams worldwide; however, it also entailed new solutions and digital innovation in connectedness and communication, which might benefit patients and relatives.[47] A digital diary could contribute to such benefits. This new tool appeared to represent a rapid change in

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social support and connectedness for ICU patients' relatives and was found to be feasible from their perspectives. Digital innovation that takes the patient and his or her self-care into account is increasing.[48] Nurses should participate in these innovative transformations in health care as well as in other emerging areas such as the use of data science, artificial intelligence, e-Health and m-health.[49]

6.2 | Strengths and limitations

Despite the thorough design of the study and its careful implementation, several limitations need to be addressed. First, an online questionnaire method was used. Because of a high proportion of early adopters, an overestimated positive response bias might have influenced the evaluation of the digital diary. Relatives with higher levels of technophobia tend to limit interaction with digital technology as much as possible, so this group may have been underrepresented in this study. Usability and digital use of the Post-ICU diary might therefore not be perceived from the perspectives of relatives with fear from advanced technologies. Offering a paper questionnaire as an alternative could have been helpful for recruiting respondents for the study with minimal digital competencies. Therefore, the results on technophobia should be interpreted with caution. Second, the tool used in this study was investigator-developed, with items to assess the usability adapted to the new intervention. It was pretested by only three volunteers who provided feedback and asserted the tool's face validity. Pretesting by more volunteers from different background is warranted for robust development of a tool measuring feasibility. A questionnaire with 65 items seems rather lengthy; however, it took approximately 10–15 min to complete, as known from the testing phase. This length was not too challenging and all received questionnaires were completed until the end. Third, despite guaranteeing anonymity and emphasizing that all answers were 'correct', socially desirable answers cannot be ruled out. Fourth, because of the cross-sectional study design, causality cannot be established in factors related to the intention to use the Post-ICU diary. Fifth, the overall small sample size reduces the generalizability of its findings.

6.3 | Implications for practice

The implementation of this new digital tool supports a person-centred ICU policy because of its focus on the personal diary entries of the patient and the collaborative writing process featuring relatives and professionals. When implementing a digital diary, it is important to bear in mind its ease of use by regularly asking for feedback regarding its usability. ICU patients and their relatives must be included in the development of new features for a digital tool, and they must be consulted when making decisions regarding how to prioritize the release of improvements. This approach has already led to a more efficient print functionality and login procedures for the digital diary. By adapting the Post-ICU diary to participants' feedback, its PEOU can be increased further.

The results of this small-scale pilot study offer a preliminary perspective on the implementation of the Post-ICU diary in Dutch ICUs. A follow-up feasibility study is necessary to obtain insights into the effectiveness of a digital diary in reducing health-related impairments related to PICS and PICS-F in ICU patients and their relatives. Additional data regarding the symptoms of PICS/PICS-F may support the implementation of a digital diary. The digital Post-ICU diary is a valuable alternative to a paper diary, and it is recommended that at least one of these versions should be implemented in every ICU.

7 | CONCLUSION

Participants perceived a digital diary as highly valued and easily feasible. PEOU, PU and technophobia were not found to influence the usability of the digital diary. A digital diary might meet the social support and connectedness needs of ICU patients' relatives. The implementation of this innovation in the use of ICU diaries supports a person-centred policy for its focus on the personal diary entries of the patient and the collaborative writing process featuring relatives and professionals.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

PATIENT CONSENT

All participants provided signed informed consent.

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